PRESIDENCY UNIVERSITY
Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064

AI SOLUTIONS INTERN

AN INTERNSHIP REPORT

Submitted by

Ratish Gurav - 20221CSG0004

Under the guidance of,

Mr. Saptarsi Sanyal

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND TECHNOLOGY
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

GAIN MORE KNOWLEDGE
REACH GREATER HEIGHTS

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BENGALURU

APRIL 2026

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report **AI Solution Intern** is a bonafide work of **Ratish Gurav - 20221CSG0004,** who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND TECHNOLOGY (Artificial Intelligence and Machine Learning) during 2025-2026.

Mr. Saptarsi Sanyal Project Guide PSCS Presidency University	**Mr. Saptarsi Sanyal** Program Internship Coordinator PSCS Presidency University	22/04/2026 **Dr. Md Ziaur Rahaman** School Internship Coordinator PSCS Presidency University
22/04/26 **Dr. Pallavi R** Head of the Department, PSCS Presidency University	**Dr. Duraipandian N** Dean PSCS / PSIS Presidency University	

Name and Signature of the Examiners

Sl. No	Name	Signature	Date
1	Saptarsi Sanyal		23/04/26
2.	Ms. Akshara Sinah Bijn		23/04/26

PRESIDENCY UNIVERSITY

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DECLARATION

I am a student of final year B.Tech in COMPUTER SCIENCE AND TECHNOLOGY (Artificial Intelligence and Machine Learning), at Presidency University, Bengaluru, named **Ratish Gurav** hereby declare that the internship work titled **AI Solutions Intern** has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND TECHNOLOGY (Artificial Intelligence and Machine Learning) during the academic year of 2025-2026. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Ratish Gurav USN: 20221CSG0004 Signature

PLACE: BENGALURU

DATE: 20 - April 2026

INTERNSHIP COMPLETION CERTIFICATE

CERTIFICATE

OF INTERNSHIP

Proudly Awarded To:

Ratish Gurav

This is to certify that Mr. Ratish Gurav has successfully completed his internship as part of his academic requirements. The work demonstrates good technical skills and the ability to apply knowledge to real-world problems. We wish him success in his future endeavors.

Internship Period: 15th January 2026 – 15th April 2026

RAJEEV GURAV
Director

ATMARAM BORDEKAR
Project Lead

ACKNOWLEDGEMENTS

For completing this internship work, I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. I extend my gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the internship.

I would like to sincerely thank my internal guide **Mr. Saptarsi Sanyal**, Presidency School of Computer Science and Engineering, Presidency University, for her moral support, motivation, timely guidance and encouragement provided to me during the period of internship work.

I am also thankful to **Dr. Pallavi R, Professor, Head of the Department, Presidency School of Computer Science and Engineering** Presidency University, for her mentorship and encouragement.

I express my cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my internship work.

I am grateful to **Dr. Md Ziaur Rahaman,** Internship Coordinator and **Mr. Saptarsi Sanyal** Program Internship Coordinator, Presidency School of Computer Science and Engineering, or facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

I am also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

Ratish Gurav

ABSTRACT

The rapid growth of sustainable energy solutions such as solar panels, electric vehicle charging systems, and energy-efficient infrastructure has significantly increased the demand for technical consultation and customer support services. Organizations operating in this domain often face challenges in handling large volumes of customer queries related to product specifications, installation procedures, and performance optimization. Traditional customer support systems rely heavily on manual intervention, leading to delays, inefficiencies, and scalability issues.

This internship focuses on the design and development of an AI-powered chatbot system aimed at automating technical consultation and improving customer interaction. The system is built using a Retrieval-Augmented Generation (RAG) architecture, which combines the capabilities of Large Language Models (LLMs) with a document retrieval mechanism to ensure accurate and context-aware responses. The frontend of the system is developed using React.js to provide a seamless user interface, while the backend is implemented using FastAPI to handle asynchronous requests and integrate with AI services. A vector database is utilized to store embeddings of technical documents, enabling efficient retrieval of relevant information.

In addition to technical development, the internship also involved active participation in client meetings and business discussions, providing insights into real-world requirements and customer expectations. Responsibilities included tracking client requirements, understanding business workflows, and assisting in aligning technical solutions with organizational goals.

The developed system demonstrates improved response accuracy, reduced dependency on manual support, and enhanced user experience. The project highlights the potential of integrating AI-driven solutions in traditional industries to achieve scalability, efficiency, and digital transformation.